

# Target validation paths — hgsoc-tuboovarian-stage3c-r0-hrd-pen ding-idyq

*The diagnostic and biomarker workup that hardens the targetable-feature call*

Generated 2026-06-24

# Target validation paths

One pending result decides the maintenance strategy: the tumor HRD genomic-instability score plus somatic BRCA1/2, run on archival FFPE from the interval-debulking specimen. A positive read (Myriad MyChoice GIS at or above 42, FoundationOne CDx genomic LOH-high, or a somatic BRCA1/2 mutation) qualifies olaparib plus bevacizumab via NCT02477644 and, when a somatic BRCA mutation is present, single-agent olaparib via NCT01844986. niraparib via NCT02655016 is an option regardless of the score, with larger benefit if it returns positive. If the genomic-instability score is negative and somatic BRCA is wild-type, the two HRD-positive PARP options drop out; the next conversation about standard 2L+ care belongs to the treating team, not this report. The remaining assays below sharpen context and later-line sequencing rather than gating the frontline choice.

## HRD

The essential test is the tumor HRD genomic-instability score plus somatic BRCA1/2 on archival FFPE, ordered as an FDA companion diagnostic (Myriad MyChoice CDx at the GIS at-or-above-42 cutoff, or FoundationOne CDx reporting genomic LOH). One assay returns both pending biomarkers, so a single FFPE order resolves the somatic BRCA call and the genomic-instability score together. Germline BRCA is already wild-type, so this covers the somatic and genomic-scar gap germline testing cannot reach. Two high-priority co-tests should run on the same block: MSI / MMR status (MMR IHC for MLH1, PMS2, MSH2, MSH6, or MSI by NGS or PCR), since dMMR or MSI-high would open tumor-agnostic checkpoint-inhibitor eligibility, and a comprehensive somatic NGS panel that captures non-BRCA HR-pathway genes (RAD51C, RAD51D, PALB2, BRIP1) plus other actionable drivers. As a low-priority, later option, a plasma ctDNA panel for BRCA1/2 reversion variants can monitor for acquired PARP-inhibitor resistance once maintenance is underway, but it does not gate the baseline decision.

## ER

ER is already reported positive at about 80% of nuclei with weak-to-moderate intensity, which supports a low-toxicity endocrine-maintenance rationale that fits her age and preference for oral therapy. A reference-laboratory ER / PR IHC read with standardized ASCO/CAP scoring (Allred or percent-plus-intensity) is medium priority: it confirms staining quality and quantifies intensity, which matters because PR-negativity tempers the expected endocrine benefit. This refines later-line sequencing rather than gating the frontline maintenance decision.

## IL-12

The patient is on investigational IL-12 immunogene therapy whose mechanism depends on a T-cell-inflamed microenvironment. Tumor-infiltrating lymphocyte density (CD3/CD8) and PD-L1 IHC (22C3 or SP263), with multiplex immune profiling where available, characterize that context. These are low-priority, research-grade reads for the on-study setting; they do not gate a standard-of-care decision and are best treated as supportive correlative data.

## Where to order these assays

| Assay | Provider | Decision gated | Contact |
|-------|----------|----------------|---------|
|-------|----------|----------------|---------|

|                                                                                                                                                                     |                                                                   |                                                                                                                                                               |                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Tumor HRD genomic-instability score plus somatic BRCA1/2 (FDA companion diagnostic; e.g. Myriad MyChoice CDx, GIS &gt;=42, or FoundationOne CDx genomic LOH)</b> | <b>Myriad Genetics (preferred) (MyChoice CDx)</b>                 | <b>Olaparib + bevacizumab maintenance (PAOLA-1) if HRD-positive; niraparib maintenance regardless of HRD, with magnitude of benefit refined by the score.</b> | <b>test info · 322 North 2200 West, Salt Lake City, UT 84116 · 1-800-469-7423</b>  |
| Tumor HRD genomic-instability score plus somatic BRCA1/2 (FDA companion diagnostic; e.g. Myriad MyChoice CDx, GIS >=42, or FoundationOne CDx genomic LOH)           | Foundation Medicine (FoundationOne CDx)                           | Olaparib + bevacizumab maintenance (PAOLA-1) if HRD-positive; niraparib maintenance regardless of HRD, with magnitude of benefit refined by the score.        | test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639                |
| Tumor HRD genomic-instability score plus somatic BRCA1/2 (FDA companion diagnostic; e.g. Myriad MyChoice CDx, GIS >=42, or FoundationOne CDx genomic LOH)           | Caris Life Sciences (MI Cancer Seek / Genomic LOH)                | Olaparib + bevacizumab maintenance (PAOLA-1) if HRD-positive; niraparib maintenance regardless of HRD, with magnitude of benefit refined by the score.        | test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669              |
| Tumor HRD genomic-instability score plus somatic BRCA1/2 (FDA companion diagnostic; e.g. Myriad MyChoice CDx, GIS >=42, or FoundationOne CDx genomic LOH)           | Tempus (Tempus xT / HRD)                                          | Olaparib + bevacizumab maintenance (PAOLA-1) if HRD-positive; niraparib maintenance regardless of HRD, with magnitude of benefit refined by the score.        | test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137 |
| <b>MSI / MMR status (MMR IHC for MLH1, PMS2, MSH2, MSH6 and/or MSI by NGS or PCR)</b>                                                                               | <b>NeoGenomics Laboratories (preferred) (MMR IHC panel / MSI)</b> | <b>Tumor-agnostic checkpoint inhibitor (pembrolizumab / dostarlimab) eligibility in dMMR / MSI-high disease.</b>                                              | <b>test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907</b>     |
| MSI / MMR status (MMR IHC for MLH1, PMS2, MSH2, MSH6 and/or MSI by NGS or PCR)                                                                                      | Quest Diagnostics (MMR IHC / MSI by PCR)                          | Tumor-agnostic checkpoint inhibitor (pembrolizumab / dostarlimab) eligibility in dMMR / MSI-high disease.                                                     | test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378                   |
| MSI / MMR status (MMR IHC for MLH1, PMS2, MSH2, MSH6 and/or MSI by NGS or PCR)                                                                                      | Caris Life Sciences (MSI by NGS)                                  | Tumor-agnostic checkpoint inhibitor (pembrolizumab / dostarlimab) eligibility in dMMR / MSI-high disease.                                                     | test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669              |
| <b>Comprehensive somatic NGS panel (BRCA1/2, RAD51C/D, PALB2, BRIP1, other HR-pathway genes; plus broad actionable drivers and tumor mutational burden)</b>         | <b>Foundation Medicine (preferred) (FoundationOne CDx)</b>        | <b>Identifies non-BRCA HR-pathway alterations supporting PARP-inhibitor use and screens for any off-pathway actionable driver.</b>                            | <b>test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639</b>         |
| Comprehensive somatic NGS panel (BRCA1/2, RAD51C/D, PALB2, BRIP1, other HR-pathway genes; plus broad actionable drivers and tumor mutational burden)                | Tempus (Tempus xT)                                                | Identifies non-BRCA HR-pathway alterations supporting PARP-inhibitor use and screens for any off-pathway actionable driver.                                   | test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137 |

|                                                                                                                                                      |                                                                         |                                                                                                                             |                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Comprehensive somatic NGS panel (BRCA1/2, RAD51C/D, PALB2, BRIP1, other HR-pathway genes; plus broad actionable drivers and tumor mutational burden) | Caris Life Sciences (MI Cancer Seek)                                    | Identifies non-BRCA HR-pathway alterations supporting PARP-inhibitor use and screens for any off-pathway actionable driver. | test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669              |
| <b>ER / PR IHC, reference-laboratory confirmation with ASCO/CAP scoring (Allred or percent-plus-intensity)</b>                                       | <b>NeoGenomics Laboratories (preferred) (ER/PR IHC)</b>                 | <b>Endocrine (aromatase-inhibitor) maintenance or later-line rationale in ER-positive HGSC.</b>                             | <b>test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907</b>     |
| ER / PR IHC, reference-laboratory confirmation with ASCO/CAP scoring (Allred or percent-plus-intensity)                                              | Quest Diagnostics (ER/PR by IHC)                                        | Endocrine (aromatase-inhibitor) maintenance or later-line rationale in ER-positive HGSC.                                    | test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378                   |
| <b>Plasma ctDNA panel for BRCA1/2 reversion and HR-pathway resistance variants</b>                                                                   | <b>Guardant Health (preferred) (Guardant360 CDx)</b>                    | <b>PARP-inhibitor resistance surveillance via acquired BRCA reversion; does not gate baseline maintenance selection.</b>    | <b>test info · 3100 Hanover Street, Palo Alto, CA 94304 · 1-855-698-8887</b>       |
| Plasma ctDNA panel for BRCA1/2 reversion and HR-pathway resistance variants                                                                          | Foundation Medicine (FoundationOne Liquid CDx)                          | PARP-inhibitor resistance surveillance via acquired BRCA reversion; does not gate baseline maintenance selection.           | test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639                |
| <b>Tumor-infiltrating lymphocyte density and PD-L1 IHC (CD3/CD8 plus PD-L1 22C3 or SP263), with multiplex immune profiling if available</b>          | <b>NeoGenomics Laboratories (preferred) (PD-L1 IHC / TIL profiling)</b> | <b>Correlative immune context for on-study IL-12 therapy; does not gate standard-of-care selection.</b>                     | <b>test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907</b>     |
| Tumor-infiltrating lymphocyte density and PD-L1 IHC (CD3/CD8 plus PD-L1 22C3 or SP263), with multiplex immune profiling if available                 | Tempus (Tempus xT / immune profiling)                                   | Correlative immune context for on-study IL-12 therapy; does not gate standard-of-care selection.                            | test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137 |

## Biomarker plan

| Recommended assay | Rationale | Suggested assay provider | Tissue requirements |
|-------------------|-----------|--------------------------|---------------------|
|-------------------|-----------|--------------------------|---------------------|

|                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                    |                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <p>Tumor HRD genomic-instability score plus somatic BRCA1/2 (FDA companion diagnostic; e.g. Myriad MyChoice CDx, GIS &gt;=42, or FoundationOne CDx genomic LOH)</p> | <p>This pending result is the primary decision point for a newly diagnosed, platinum-responsive, R0 Stage IIIC high-grade serous case with wild-type germline BRCA. A positive HRD score (Myriad MyChoice GIS &gt;=42 or a somatic BRCA1/2 mutation) qualifies the patient for olaparib plus bevacizumab maintenance per PAOLA-1, while niraparib maintenance is an option regardless of HRD but with larger benefit when HRD-positive. Without this assay the maintenance choice cannot be made correctly; the deep platinum and ctDNA response is consistent with HRD but is not diagnostic of it.</p> | <p>Myriad Genetics (MyChoice CDx) · test info · 322 North 2200 West, Salt Lake City, UT 84116 · 1-800-469-7423</p> | <p>archival FFPE acceptable (no fresh biopsy needed)</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|

|                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                 |                                 |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <p>MSI / MMR status (MMR IHC for MLH1, PMS2, MSH2, MSH6 and/or MSI by NGS or PCR)</p> | <p>MSI-high or mismatch-repair-deficient status would open tumor-agnostic checkpoint-inhibitor eligibility (pembrolizumab or dostarlimab), an option the current profile cannot rule in or out because the result is not back. dMMR is uncommon in high-grade serous ovarian cancer but is inexpensive to exclude and changes later-line strategy if present. MMR IHC and MSI usually run on the same FFPE block already submitted for the HRD panel.</p> | <p>NeoGenomics Laboratories (MMR IHC panel / MSI) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907</p> | <p>archival FFPE acceptable</p> |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------|

|                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                |                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------|
| Comprehensive somatic NGS panel (BRCA1/2, RAD51C/D, PALB2, BRIP1, other HR-pathway genes; plus broad actionable drivers and tumor mutational burden) | A comprehensive panel captures non-BRCA genes (RAD51C, RAD51D, PALB2, BRIP1) that contribute to HRD and to PARP-inhibitor benefit beyond the genomic-instability score alone, and it flags any other actionable driver or high tumor mutational burden. Restricting tumor sequencing to BRCA1/2 would miss HR-pathway alterations and any off-pathway target. This is usually the same panel that returns the somatic BRCA and HRD score, so it adds breadth at no extra tissue cost.                                                                  | Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639  | archival FFPE acceptable |
| ER / PR IHC, confirmation with ASCO/CAP scoring (Allred or                                                                                           | ER is already reported positive at about 80% of nuclei with weak-to-moderate intensity, which supports a low-toxicity endocrine-maintenance rationale (aromatase inhibition) that fits the patient's age and preference for oral, low-burden therapy. A reference-laboratory ER/PR read with standardized scoring confirms the staining quality and quantifies intensity, since PR-negativity tempers the expected magnitude of endocrine benefit. This refines later-line sequencing rather than gating a frontline decision, so it is not essential. | NeoGenomics Laboratories (ER/PR IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907 | archival FFPE acceptable |

|                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                |                              |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Plasma ctDNA panel for BRCA1/2 reversion and HR-pathway resistance variants                                                          | <p>If a somatic BRCA1/2 mutation is confirmed and PARP-inhibitor maintenance begins, plasma ctDNA can later detect secondary reversion mutations that restore homologous recombination and drive PARP-inhibitor resistance. This is a surveillance tool for the maintenance phase, not a baseline gate, so it is low priority now; the patient already has tumor-informed Signatera ctDNA that tracks residual disease. Order it only at biochemical or molecular progression on a PARP inhibitor.</p> | Guardant Health (Guardant360 CDx) · test info · 3100 Hanover Street, Palo Alto, CA 94304 · 1-855-698-8887                      | two 10 mL Streck blood tubes |
| Tumor-infiltrating lymphocyte density and PD-L1 IHC (CD3/CD8 plus PD-L1 22C3 or SP263), with multiplex immune profiling if available | <p>The patient is on investigational IL-12 immunogene therapy (GEN-1 / IMNN-01) on trial, whose mechanism depends on a T-cell-inflamed microenvironment. CD3/CD8 infiltrate density and PD-L1 status characterize that context and inform any reasoning, but they are research-grade for this on-study setting and do not gate a standard-of-care decision. Treat as supportive correlative data, not part of the required workup.</p>                                                                 | NeoGenomics Laboratories (PD-L1 IHC / TIL profiling) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907 | archival FFPE acceptable     |

*Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.*